

Padlet links

All Tufts Made Padlets

https://docs.google.com/presentation/d/1ES9oZh56kIV36pI9BEHiGUiqgh95-u5D/edit?slide=id.g2d8602d18f9_0_735#slide=id.g2d8602d18f9_0_735

The screenshot shows a Google Slides presentation titled "BAA Plant and Animal Resources (Padlet)". The interface includes a top menu bar with "File", "Edit", "View", "Help", and "Accessibility". To the right of the menu are icons for undo, redo, star, and share, along with a "Share" button. Below the menu is a toolbar with search, print, and zoom controls, and a "View only" button. On the left side, there is a vertical list of slide thumbnails numbered 1 through 6. The main slide area displays a large, stylized notepad with the title "BAA Padlet Resources" in bold orange text. A yellow sticky note on the notepad reads "Biomimicry Inspiration". The background of the slide is a light blue pattern of small green leaves. The thumbnails on the left show various content: slide 1 is the title slide, slide 2 says "To see these resources on a Padlet site..", slide 3 is titled "Flinging", slide 4 is titled "Click Beetle" and includes an image of a beetle, slide 5 is titled "Click Beetle" and includes a diagram of a beetle's leg, and slide 6 is titled "Click Beetle" and includes a diagram of a beetle's leg.

Digging

<https://padlet.com/TuftsCEE0/digging-370winala6hbvhm2>

Padlet

TuftsCEE0 • 3mo

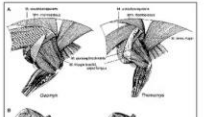
Digging

Search Slideshow Share Sign up for Padlet

Pocket Gopher (Scratch Digging)

Scratch Digging (Pocket Gopher)

not claws of any species (Figure 1).



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Copy of Pocket Gophers

Eastern or Common Mole (Humeral Rotation Digging)

Humeral Rotation Digging (Eastern or Common Mole)



Figure 1. Eastern mole.

eastern mole is highly adapted for digging. It has very large, hairless, spade-shaped front feet that are much broader than they are long (Figure 1). The bones in the hand are enlarged to support the weight of the front feet, which are usually held in a vertical position. The front feet are also enlarged and provide strong support for the body when the mole is digging.

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Copy of Eastern or Common Mole

Mole Rats and Pocket Gophers (Chisel-tooth Digging)

Chisel-tooth Digging (Mole Rats and Pocket Gophers)



Figure 1. Chisel-tooth digging.

Chisel-tooth digging is a type of digging that uses the front teeth to break up soil. Mole rats and pocket gophers are examples of animals that use this method.

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Copy of Mole Rats and Pocket Gophers

Pangolin (Hook-and-pull Digging)

Hook-and-Pull Digging (Pangolins)



Figure 1. Hook-and-pull digging.

Pangolins have extremely large and elongated claws on their fore and hind feet (Figure 1). They use these claws to dig into the soil and pull out the insects they are eating.

google docs

Copy of Pangolins

Pocket Gopher Digging



YouTube

Gopher Digging a Hole

Mole Digging Video



YouTube

How Hard Do Moles Dig? | ScienceTake | The New York Times

Mole Rat Digging



YouTube

Digging mole-rat

Pangolin Digging



YouTube

Pangolin | World's Weirdest

Gopher Digging



Mole Video - Shortened



Mole Rat Video - Shortened



Pangolin Digging



Flinging

<https://padlet.com/TuftsCEE0/flinging-lca9qyr08c6j74bm>

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Flinging

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Seed Pods



Seed Pods - Google Docs

Seed pods are adapted to fling their seeds into the air. This self-launching process, called "mechanical dispersal," is driven by physical forces rather than by animals that eat the seeds. The pods for vetches (Figure 1) and lupines (Figure 2) have different structures that rely on the forces of dispersing seeds. However, these different structures work the same way: they both fling their seeds by changing the shape of their seed pods.



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Copy of Seed Dispersal Student-Facing Handout

Click Beetle



Click Beetle - Google Docs



Figure 1: Click Beetle

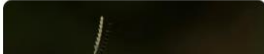
How does the click beetle make such impressive jumps? It all comes down to a special bit of anatomy that can store and release energy at the center of the beetle's body. Located between the elytra, a structure called the spring consists of a hinge with two pegs. Each peg fits into a slot in the elytra (Figure 2).

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Copy of Click Beetle Student-Facing Handout

Additional Resources

Energy Stored to Amplify Power

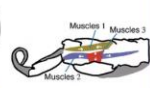


Spring Tail



Spring Tail - Google Docs

As the upper muscles (shaded yellow in Figure 1 and 2) are extended (in their long position), the lower muscles (shaded purple in Figure 2) are contracted (in their short position). Then, the tail acts as a lever (Figure 2, left). The muscles are storing up energy for the jump. When the upper muscles contract (shorten) and the lower muscles extend (lengthen), a catapult is formed (Figure 2, right). The energy stored in the muscle "spring" gets transferred to the rest of the springtail.



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Copy of Springtail Student Facing Handout

Additional Resources

Appendage Causes High Jumps



Energy Absorption

<https://padlet.com/TuftsCEE0/energy-absorb-rk5doxda7izskizh>

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Energy Absorb

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Pomelo Fruit

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2 years ago

Pomelo Fruit



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Pomelo Fruit

Hedgehog

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Hedgehog



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Hedgehog

Scaly-Foot Snail

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Scaly-Foot Snail



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Scaly Foot Snail

Woodpecker

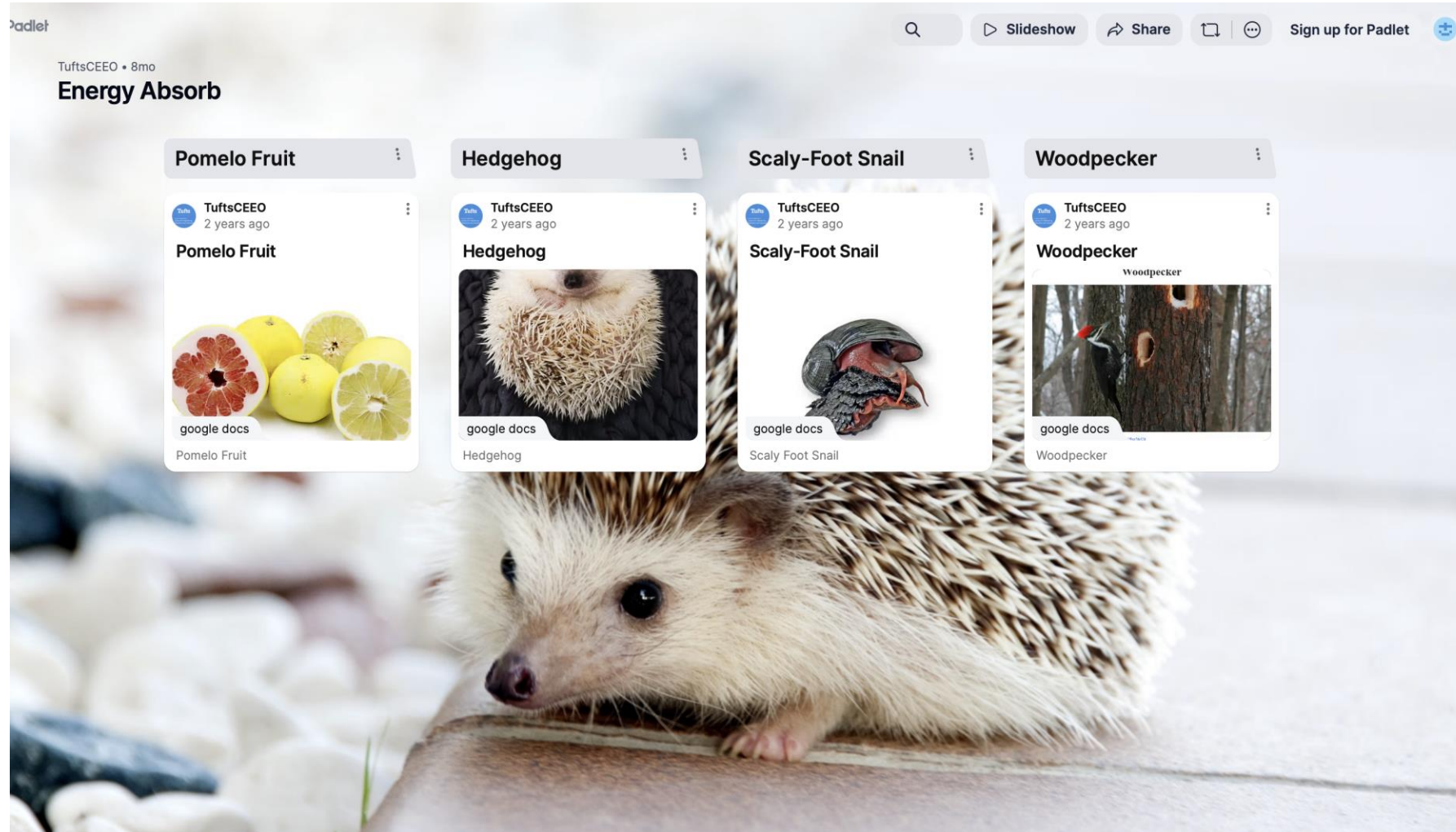
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2 years ago

Woodpecker



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Woodpecker



Wind Dispersal

<https://padlet.com/TuftsCEE0/wind-dispersal-z5g2eu1x37xq4p1b>

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Wind Dispersal

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Anisoptera Tree Seeds

TuftsCEE0 a year ago

Anisoptera Tree Seeds

As anisoptera tree seeds are unequal in length. This asymmetry structure causes the seeds to spin v. / are released from the mother trees and enables long-distance dispersal. When the wind /s, the seeds look like a tiny whirling helicopter, which may travel even further than a ble-winged glider.



google docs

Copy of Anisoptera tree seeds

Dandelion Seeds

TuftsCEE0 a year ago

Dandelion Seeds

A dandelion seed looks like a bird. The seed head opens into a parachute-like structure when the seed is ready. Each dandelion seed is attached by a thin tube to around 100 bristles, forming this white-like structure. When the seeds detach from the flower head, these bundles of hairs in the wind and carry the seeds away. As the seed head dries, it opens up, and each seed has a parachute that allows it to float and fly away even in the lightest breeze. This hairy white closes when the air is humid, which often means the wind is weak.



google docs

Copy of Dandelion seeds

Maple Tree Seeds

TuftsCEE0 a year ago

Maple Tree Seeds

As they fall to the ground and helps the maple trees disperse their seeds over a wider area.



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Copy of Maple tree seed pods

Pine Tree Pollen Grain

TuftsCEE0 a year ago

Pine Tree Pollen Grain



Copy of Pine Tree Pollen Grains

Clematis Seed Heads

TuftsCEE0 8 months ago

Clematis Seed Heads



Copy of Clematis Seed Heads

Grasping

<https://padlet.com/TuftsCEE0/grasping-8aqcinra3exqb8tm>

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Grasping

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Bird Talons



Bird Talons - Google Docs



Phalanx

sesamoid bone

unguitractor bone

Copy of Bird Talons

Venus Fly Traps



Venus Fly Traps - Google Docs



Trigger hairs

digestive glands

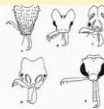
central cavity

Copy of Venus Fly Trap

Trap-Jaw Ant



Trap-Jaw Ant - Google Docs



Mandibles

Trigger hairs

central cavity

Copy of Trap Jaw Ant

Additional Resources

Why Birds Can Sleep on Branches and Not Fall Off

Copy of Why Birds Can Sleep on Branches and Not Fall Off

Additional Resources

Trap-Jaw Ants

Copy of Trap-Jaw Ants

Leaping – Beth Travers

<https://padlet.com/bethtravers/leaping-organisms-qc1pk4ubsal8qm1s>

Padlet

Beth Travers • 6mo

Leaping Organisms

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High Jumping Spring Tail

Beth Travers 6 months ago



asknature.org

Appendage Causes High Jumps - Biological Strategy - AskNature

Geared Legged Insect

Beth Travers 6 months ago



smithsonianmag.com

This Insect Has The Only Mechanical Gears Ever Found in Nature

Water Jumping Cricket

Beth Travers 6 months ago



sciencedirect.com

ScienceDirect

Hopping Grasshopper

Beth Travers 6 months ago



st-andrews.ac.uk

How the legs work

Springtails

Beth Travers 6 months ago



YouTube

Springtails are INCREDIBLE! What are they & why are they important?

Geared-up Insects

Beth Travers 6 months ago



sdnhm.org

Geared-up Insects Put the Mechanics into Biomechanics

The Company of Biologists

Beth Travers 6 months ago



prelights.biologists.com

The physiological adaptation for the "fore-mid" four-legged walking behavior of the pygmy mole cricket Xya sichuanensis - preLights

Grasshoppers

Beth Travers 6 months ago



asknature.org

Grasshoppers Store Energy to Power Super Leaps - Biological Strategy - AskNature

Flipping – Beth Travers

<https://padlet.com/bethtravers/flipping-organisms-493uzk2iy47cxy3t>

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Flipping Organisms

Click Beetle

YouTube

How do click beetles jump?

Spring Tail

YouTube

Springtails do the fastest backflip on earth!

Horseshow Crabs

YouTube

Horseshoe Crab Turns Over

Domed Tortoise

YouTube

Getting back on your feet; the Galapagos Giant tortoise way

Locusts

interestingengineering.com

Scientists Study Incredible Energy of Click Beetles' Jump Motion

snexplores.org

A springtail's spring-loaded backflip is fast - really fast

dnr.sc.gov

SCDNR - Horseshoe Crabs

discovermagazine.com

Shell Shape Helps Tortoises Get Up

Other links

- <https://asknature.org/>
- <https://www.ngssphenomena.com/>

